

P-V Hysteresis as a Fatigue Leading Indicator and Its Degradation of Pressure-Only Proprioception in Shared-Manifold Soft Grippers

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Abstract

Soft pneumatic grippers present two concurrent monitoring challenges: detecting actuator fatigue before rupture and reconstructing gripper pose from existing pressure signals without additional sensors. Prior work addresses each in isolation. We show that a shared pneumatic manifold — the most practical topology for multi-finger grippers — creates a third, previously uncharacterized challenge: inter-chamber pressure cross-talk couples both problems. We fabricate a three-chamber silicone gripper on a shared manifold and make two primary contributions. First, we demonstrate that pressure-volume (P-V) hysteresis loop shape is a reliable leading indicator of silicone actuator fatigue, with measurable loop-shape change detectable [X] cycles before mechanical rupture (N=10). Second, we show that shared-manifold cross-talk degrades pressure-only pose reconstruction accuracy from [A]% to [B]%, and that a cross-talk correction model recovers accuracy to within [C]% of the isolated-chamber baseline. We further characterize how fatigue-induced P-V drift and proprioceptive accuracy degradation are coupled: as an actuator ages, the P-V signature shift that signals fatigue also shifts the cross-talk correction model, causing both to fail simultaneously. Joint monitoring from the same pressure stream is therefore necessary and sufficient for health-aware soft gripper autonomy.

1. Introduction

Soft pneumatic actuators are increasingly deployed in grippers, medical devices, and wearable systems due to their compliance and inherent safety. Two monitoring problems arise in practice. The first is **health monitoring**: silicone actuators fail by cyclic fatigue, but current practice detects failure only after rupture — a brittle, unrecoverable event. The second is **proprioception**: knowing the gripper's pose without expensive external sensors is essential for closed-loop control.

Recent work showed that control-loop pressure readings can reconstruct pose in isolated single-chamber or independently-driven multi-chamber designs [Wang et al., *Soft Robotics* 10(4) 2023, DOI 10.1089/soro.2021.0056; Wang et al., *Adv. Intell. Syst.* 7(4) 2025, arXiv:2411.07309]. However, these designs assume each chamber's pressure is independently controllable. In practice, multi-finger grippers share a pneumatic manifold to reduce valve count and system complexity — a coupling these prior works do not address.

We identify a compounded gap. No prior work (1) uses the in-loop P-V hysteresis signature as a fatigue *leading indicator*, (2) characterizes how shared-manifold cross-talk affects pressure-only proprioception, or (3) shows that these two phenomena are coupled. Roels et al. (*Adv. Intell. Syst.* 2026, DOI 10.1002/aisy.202500699) standardized elastomer characterization but explicitly excluded fatigue; Torzini et al. (2024, DOI 10.1007/s00170-024-14216-0) and Libby et al. (2022, arXiv:2212.03420) report only

cycles-to-failure or FEM deviation — neither uses P-V loop shape as a health signal.

This paper makes three contributions:

- (1) A cyclic P-V signature dataset for Dragon Skin 20A silicone actuators showing statistically measurable loop-shape change $[X]$ cycles before rupture ($N=10$ actuators).
- (2) A cross-talk characterization and correction method for three-chamber shared-manifold grippers, recovering pose reconstruction accuracy from $[B]\%$ to $[C]\%$ (isolated baseline: $[A]\%$).
- (3) A joint health-and-pose monitoring framework demonstrating that fatigue and cross-talk failure modes are coupled and must be tracked from the same pressure stream.

2. Background and Related Work

2.1 Soft Actuator Fatigue

Cyclic fatigue in soft pneumatic actuators is under-characterized relative to rigid-body systems. Roels et al. (2026) provide a standardized framework for elastomer property measurement but explicitly scope out fatigue. Existing fatigue studies report only cycles-to-failure counts or FEM-predicted pressure deviation at end-of-life. No prior work extracts loop-shape features from in-loop P-V data as a health signal, analogous to acoustic emission in metal fatigue monitoring.

2.2 Pressure-Only Proprioception

Wang et al. (*Soft Robotics* 2023) used unified soft-body encoding with an RNN to reconstruct pose from internal pressure on a single-chamber finger. Wang et al. (*Adv. Intell. Syst.* 2025) demonstrated physical reservoir computing on a fabric arm driven by internal pressures alone. Both assume isolated, independently driven chambers. The shared-manifold cross-talk condition — where actuating valve i changes pressure in chamber j — has not been studied, and constitutes the remaining gap in pressure-only proprioception.

3. Hardware Design

3.1 Gripper Fabrication

Three-chamber Dragon Skin 20A silicone gripper cast in a $[dimensions]$ mold using standard two-part molding. Chambers share a single brass manifold block with one solenoid valve per chamber (Parker or equivalent). One Honeywell HSC series pressure sensor ($\pm 0.5\%$ FS, 0–100 kPa range) is tee-fitted per chamber. All sensors sampled at 100 Hz via a microcontroller (Teensy 4.1 or equivalent).

3.2 Pressure Cycling Jig

A pneumatic cycling jig inflates and deflates each actuator between 0 and $[P_{\max}]$ kPa at $[f]$ Hz via a solenoid valve and regulated supply. P-V loops are recorded continuously; end-of-life is defined as $>20\%$ pressure drop within a single cycle.

3.3 Motion Capture Ground Truth

A 6-marker OptiTrack (or Vicon) setup records gripper fingertip pose at 120 Hz. Hardware trigger synchronizes pose and pressure streams to <1 ms jitter.

4. P-V Fatigue Signature Study

4.1 Protocol

$N=10$ Dragon Skin actuators are cycled to failure. P-V loops are sampled in 50-cycle blocks. Five features are extracted per block:

- **Loop area** (hysteresis energy, J)
- **Peak pressure** at full inflation (kPa)
- **Pressure at 80% volume** (inflation slope indicator)
- **Loop asymmetry ratio** (inflation vs. deflation area)
- **PC1 score** from PCA on loop shape across all actuators

4.2 Degradation Model

For each feature, a logistic degradation model is fit to normalized cycle count. Lead-time is defined as the number of cycles before rupture at which the feature first exceeds 2σ from its healthy-baseline distribution. AUC of an early-warning classifier (feature threshold vs. rupture within next $[K]$ cycles) is reported.

4.3 Results (Placeholder)

Feature	Lead-time (cycles)	AUC	p-value
P-V area	TBD	TBD	TBD
Peak pressure	TBD	TBD	TBD
Pressure@80%vol	TBD	TBD	TBD
Asymmetry ratio	TBD	TBD	TBD
PC1 score	TBD	TBD	TBD

Table 1. Fatigue leading indicator comparison across five P-V features. Lead-time = cycles before rupture at which feature exceeds 2σ from baseline.

5. Cross-Talk Characterization and Correction

5.1 Cross-Talk Measurement

With chambers 2 and 3 sealed and valve 1 actuated across $[N]$ pressure levels ($0-[P_{\max}]$ kPa), pressure changes in chambers 2 and 3 are recorded. This is repeated for all three chamber pairs, yielding a 3×3 cross-talk coupling matrix at each pressure level.

5.2 Correction Models

Three models are compared for pose reconstruction from all three chamber pressures under shared-manifold coupling:

- (1) **Uncorrected baseline** — direct pressure-to-pose regression (ridge regression) with no cross-talk compensation.
- (2) **ARX-corrected** — autoregressive model with exogenous inputs subtracts estimated cross-talk component before regression.
- (3) **Echo-state network (ESN)** — reservoir of 100 nodes implicitly learns cross-talk dynamics.

5.3 Results (Placeholder)

Model	RMSE (mm)	R ²	Inference (ms)
Uncorrected baseline	TBD	TBD	TBD
ARX-corrected	TBD	TBD	TBD
Echo-state network	TBD	TBD	TBD

Table 2. Pose reconstruction accuracy under shared-manifold cross-talk. $N=2000$ pose-labeled traces per condition.

6. Coupled Failure Analysis

At $[X]\%$ of actuator life, fatigue-induced P-V drift shifts the cross-talk correction model coefficients by $[Y]\%$, degrading proprioceptive RMSE from $[C]$ mm to $[D]$ mm. A joint recalibration policy — triggered by the P-V health monitor crossing its 2σ threshold — is evaluated: recalibration restores proprioceptive RMSE to within $[E]$ mm of the fresh-actuator baseline.

This section is the core contribution distinguishing the combined study from two independent papers: the coupling is only observable when both are measured simultaneously on the same pressure stream.

7. Discussion

Scope. $N=10$ actuators, single material (Dragon Skin 20A), one gripper geometry, one manifold configuration. Results may not generalize to different silicone formulations or larger finger counts without re-characterization.

Natural extensions. Self-healing elastomers (A05 in outline) would shift the P-V signature in a way that requires re-validation of the health monitor. Fatigue-aware adaptive control — adjusting grip force as actuator life decreases — is a direct follow-on.

Comparison to external sensing. A future study should benchmark P-V-based health monitoring against visual inspection and acoustic emission as a rigorous comparison, rather than treating visual inspection as the only baseline.

8. Conclusion

We demonstrated that P-V hysteresis loop shape predicts silicone actuator fatigue $[X]$ cycles before rupture, that shared-manifold cross-talk degrades pressure-only proprioception in a characterizable and correctable way, and that the two phenomena are coupled: joint monitoring from the same pressure stream is both necessary and sufficient for health-aware soft gripper control. The contribution is not either experiment alone — both have partial prior art — but the coupled failure mode and the joint monitoring framework, which require the combined study to observe.

References

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